



CAK2HAB3 - Dasar Kecerdasan Artifisial

PENDAHULUAN

Prof. Suyanto

S1 Informatika – Fakultas Informatika

Semester Genap 2024/2025



COMPUTING

Artificial Intelligence
(B.Sc. in CMU 2018,)

Machine Learning

Deep Learning

Informatics
(Computer
Sciences)

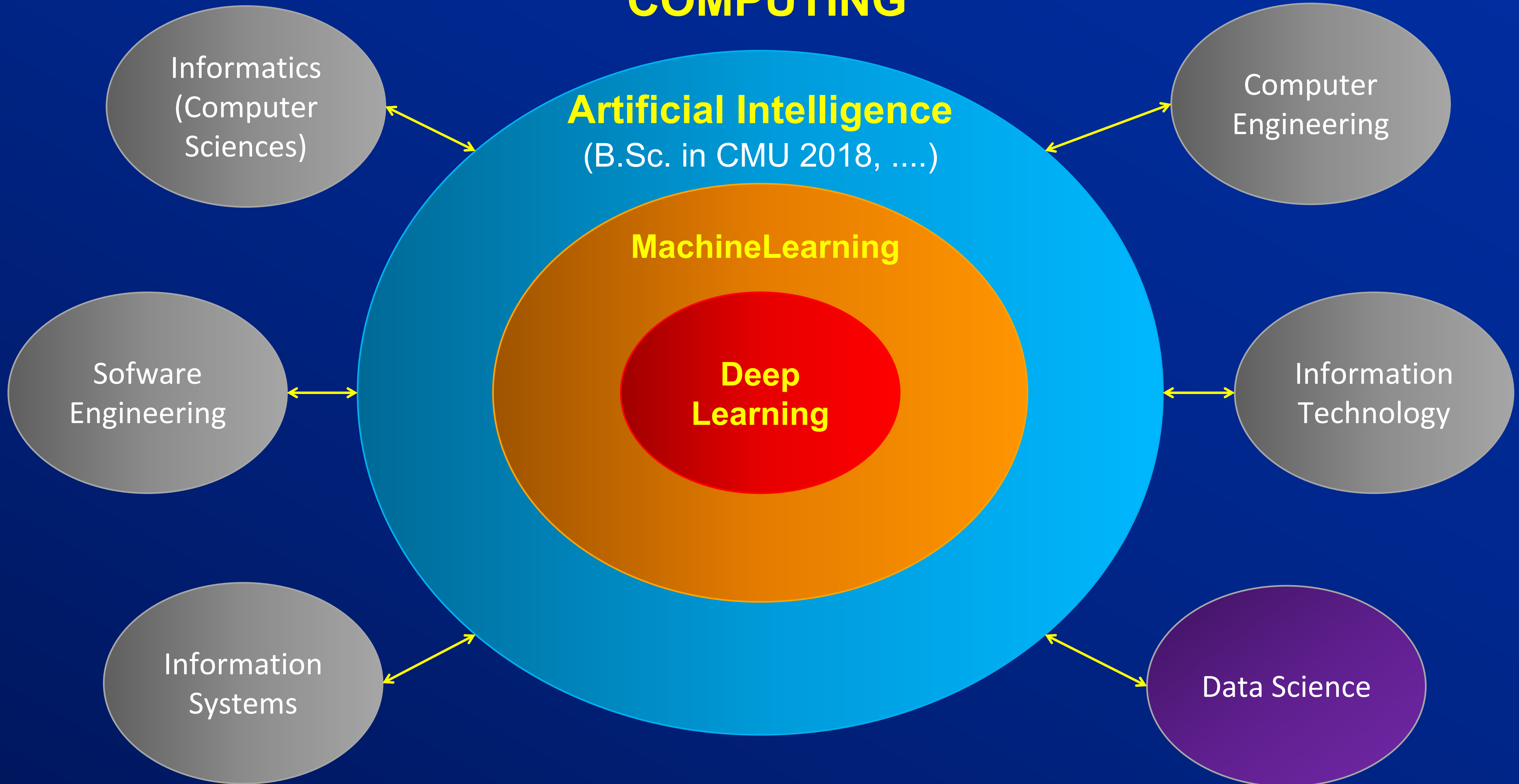
Computer
Engineering

Software
Engineering

Information
Technology

Information
Systems

Data Science



Artificial Intelligence

(Machine)
Learning

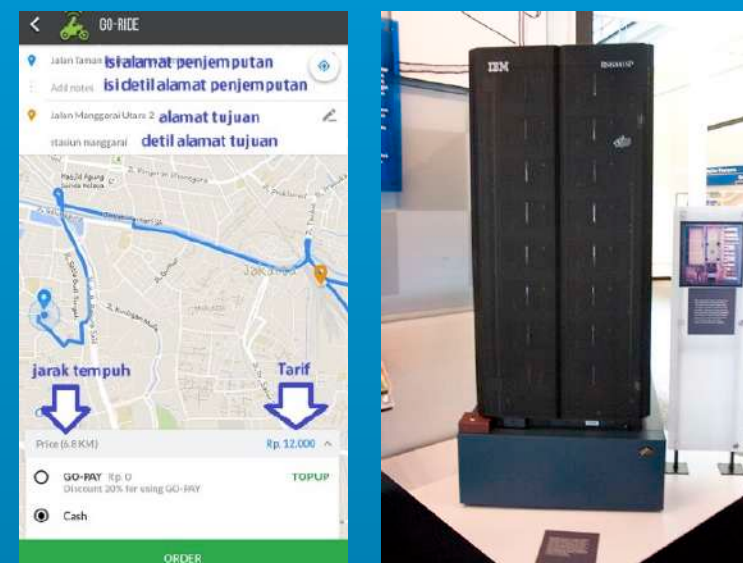
Inductive/data-driven

Deductive/rule-based

Searching

Reasoning

Planning



Machine Learning

**Deep
Learning**

**Evolutionary
Machine
Learning**

**Reinforcement
Learning**

Shallow Learning

Neural-Based
Learning

Unsupervised
Learning

Ensemble
Learning

Decision Tree
Learning

Bayesian
Learning

Support Vector
Machine

Deep Learning

**Evolutionary
Deep Learning**

**Deep
Reinforcement
Learning**

**Lifelong
Learning**

Convolutional
Neural Network

Capsule
Networks

Recurrent Neural
Network

Definisi AI

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Semester Genap 2024/2025



“The exciting new effort to make **computers think** ... machines with minds, in the full and literal sense”
(Haugeland, ‘85)

“[The automation of] activities that we associate with **human thinking**, activities such as decision making, problem solving, learning ...” (Bellman, ‘78)

Thinking Humanly

“The art of creating machines that perform functions that require **intelligence** when performed by people”
(Kurzweil, ‘90)

“The study of how to make **computers do things** which, at the moment, people do better” (Rich and Knight, ‘91).

Acting Humanly

“The study of **mental faculties** through the use of computational models”
(Charniak and McDermott, ‘85)

“The study of the computations that make it possible to **perceive, reason, and act**”
(Winston, ‘92)

Thinking Rationally

A field of study that seeks to explain and emulate **intelligent behavior** in term of computational processes” (Schalkoff, ‘90)

“The branch of computer science that is concerned with the automation of **intelligent behavior**” (Luger and Stubblefield, ‘93)

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Acting Rationally



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Sejarah AI

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COMPUTING

Artificial Intelligence (1950-an)

- Awal Perkembangan AI (1952 – 1969)
- Perkembangan AI melambat (1966 – 1974)
- Sistem berbasis pengetahuan (1969 – 1979)
- AI menjadi sebuah industri (1980 – 1988)
- Artificial Neural Networks (1986 – sekarang)

COMPUTING

Artificial Intelligence (1950-an)

Machine Learning (1980-an)

- Era Sebelum 1980
- Era 1980-an
- Era 1990 – sekarang

COMPUTING

Artificial Intelligence (1950-an)

Machine Learning (1980-an)

Deep Learning (2000-an)

- *Deep unsupervised learning*
- *Deep supervised learning*
- *Deep semi supervised learning*
- *Deep reinforcement learning*

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Aplikasi AI

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Aplikasi Berorientasi Data

- 
- Basisdata (Database)
 - Teks (Text)
 - Suara (Speech)
 - Citra (Image)
 - Audiovisual (Video)
 - Robotika (Robotic)
- 



Aplikasi dengan Basisdata

- 
- ▶ Churn Prediction (Retail, Telecommunication, etc)
 - Fraud Detection (Credit Card)
 - Recommender Systems (Ecommerce)
 - Cyber Security (Financial Technology)
 - Autonomous Networks (Telecommunication)
 - ...





Aplikasi berbasis Teks

- 
- Sentiment Analysis
 - Chatbot & Customer Support
 - Text Summarization
 - Document Clustering
 - ...





Aplikasi berbasis Suara

- 
- Automatic Speech Recognition
 - Speaker Recognition
 - Speaker Gender Classification
 - Dialect and Accent Recognition
 - Speech Emotion Recognition
 - Speech Noise Reduction/De-noising/Enhancement
 - Speech Separation
 - Spoken Language Recognition
 - ...





Aplikasi berbasis Citra

- 
- Early Cancer Detection
 - Vehicle Recognition
 - Plate Recognition
 - Gender Recognition
 - Age Classification
 - Face Recognition
 - ...





Aplikasi berbasis Video

- 
- Intelligent Video Surveillance
 - Rush Detection
 - Autonomous Car
 - ...





Aplikasi Robotika

- 
- Manufacture Robots
 - Warehousing Robots (Amazon, Alibaba)
 - Cyber-Physical Systems
 - ...





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Peluang AI

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Domain Spesifik

- 
- Aplikasi simpel tapi penting
 - Fokus pada domain tertentu





Domain Spesifik

- 
- Aplikasi simpel tapi penting
 - Fokus pada domain tertentu
 - Memberikan akurasi tinggi
 - Proses yang cepat
 - Biaya terjangkau
- 



Konten Unik

- 
- Recommender Systems: Belanja online
 - Text Summarization: Berita Nasional dan Daerah
 - Speech Recognition: Bahasa Indonesia dan Daerah
 - Text to Speech: Bahasa Indonesia dan Daerah
- 



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Tantangan AI

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Tantangan AI

- 
- Reduksi Ketidakpastian
 - Optimasi Energi
 - Generalisasi Aplikasi
 - Meyakinkan Pengguna
- 



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Masa Depan AI

S1 Informatika – Fakultas Informatika

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Pro Kontra

Kontra

- AI bisa dipakai untuk kejahatan: manipulasi berita
- AI bersifat *black-box* sehingga kurang bisa dipercaya
- AI bisa mengambil alih banyak pekerjaan manusia

Pro

- AI untuk mencegah kejahatan: deteksi berita hoax
 - Untuk tugas tertentu, performansi AI melebihi manusia
 - Masih banyak pekerjaan yang lebih baik dilakukan manusia
 - AI mampu melakukan pekerjaan berbahaya
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Kontra

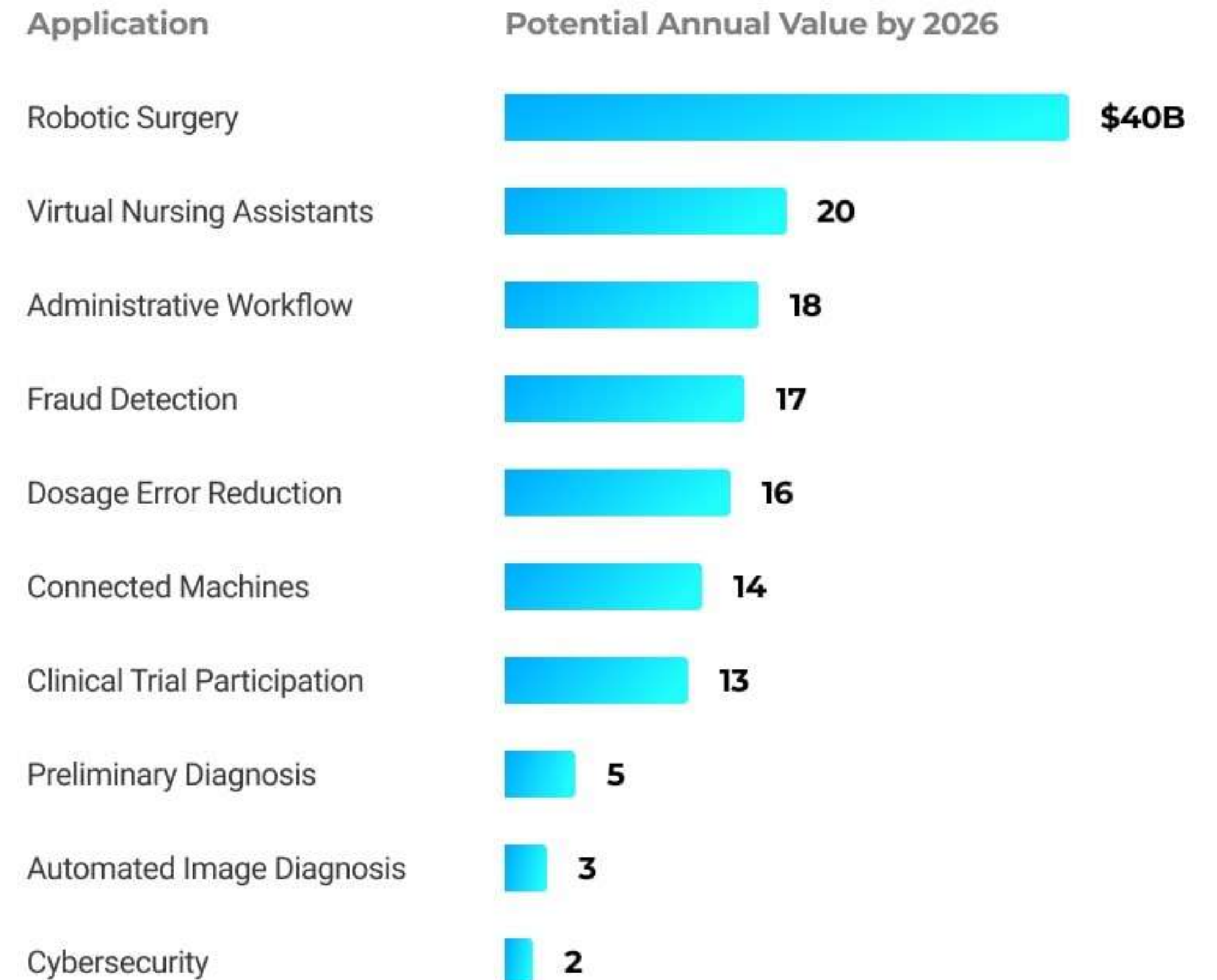
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- 

Trend Bidang AI (Teori dan Aplikasi)

- Machine Learning
- Deep Learning
- Reinforcement Learning
- Lifelong Learning
- Swarm Intelligence



Source: Accenture



Masa Depan AI

- 
- Machine Learning
 - Deep Learning





Masa Depan AI

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- Machine Learning
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Terima Kasih

